Notes: Chamley & Gale (Econometrica, 1994)
Information Revelation and Strategic Delay in a Model of Investment

Contents

1 Big picture

- **Question.** Why do agents delay investment when payoffs do not directly depend on other agents' actions, and how can delay itself prevent information from being revealed?
- **Core mechanism.** Each agent privately knows whether he has a profitable investment option. Exercising the option publicly reveals private information. Agents therefore want to wait and learn from others, but if everyone waits, no information is produced.
- **Pure informational externality.** A player's payoff depends only on his own investment decision and the true state, not directly on other players' investment. Other players matter only because their actions reveal information.
- **Equilibrium inefficiency.** The market failure must take the form of delay, investment collapse, or both. Delay is costly because investment is discounted; collapse is costly because information stops being revealed.
- **Herding/cascade connection.** When agents can react very quickly, the game ends very quickly and can generate an informational collapse resembling a herd or cascade: agents stop investing because others did not invest, not because their own information says investment is bad.
- **Period length result.** Longer reaction lags reduce the possibility of herding/collapse; very short period lengths make the process end almost instantly, but not efficiently.
- **Large-number result.** As the number of players grows, the rate of investment and information revelation become independent of the number of players. Adding more players adds more waiters, not a faster flow of information.
- **Asymmetry of mistakes.** In the large economy, surges occur only when investment is truly justified, but collapses can occur even when investment is socially optimal. The equilibrium has a bias toward underinvestment.
- **Policy intuition.** A temporary first-period investment subsidy can induce early movers, reveal information, and approximate the first best when the subsidy is small relative to the information externality.

2 Incremental contribution of the paper

- **Endogenous timing of information revelation.** Earlier cascade models often imposed the order of moves. Chamley and Gale endogenize the timing of investment, so the paper explains not only what agents learn from actions but also why some agents move first.
- **Pure-information benchmark.** The paper strips away payoff externalities, strategic complementarity, and strategic substitutability. This isolates the informational externality as the sole source of delay and collapse.
- **Unique symmetric PBE outcome.** The paper characterizes the symmetric perfect Bayesian equilibrium outcome recursively and proves uniqueness of the symmetric outcome.
- **Reaction speed as a determinant of herding.** The analysis shows that whether informational collapse occurs depends on the period length or reaction lag. This sharpens the cascade literature by showing that endogenous timing can either sustain or eliminate herd-like collapse.
- **Large-market limit.** The paper derives the limiting investment process as the number of players becomes large. The equilibrium investment rate solves an option-value condition and is independent of the number of potential investors.
- **Underinvestment bias.** The model produces a directional inefficiency: collapse can happen when investment is efficient, but a surge does not happen when investment is inefficient. This provides a theoretical foundation for recession-like coordination failure without direct payoff complementarity.

- **Policy insight.** The welfare discussion provides a rationale for temporary pump-priming: subsidize early movers to internalize the information externality and reveal the state.

3 Data and measurement: model objects and observables

This is a theoretical paper rather than an empirical paper. The "data and measurement" section therefore describes the model primitives, information objects, and observable actions that constitute the economy.

3.1 Players, options, and types

There are N agents. A random number n of them have an investment option. Agent i's type is

$$t_i = \begin{cases} 1, & \text{if agent } i \text{ has an investment option,} \\ 0, & \text{otherwise.} \end{cases}$$

The type vector is exchangeable:

$$\Pr[(t_1, \ldots, t_N)] = \Pr[(t_{\pi(1)}, \ldots, t_{\pi(N)})]$$

for any permutation π.

Interpretation. Exchangeability makes the number of option holders, not their identities, the relevant aggregate state. This permits symmetric equilibrium analysis.

3.2 Distribution of investment opportunities

The paper represents the type distribution as a two-stage experiment. First choose the total number of options n with prior probability $g_0(n)$, and then randomly allocate those options among the N agents. Conditional on having an option, an active agent's posterior distribution of n is

$$g(n) = \frac{g_0(n)n}{\sum_{n'} g_0(n')n'}.$$

Interpretation. An agent who has an option learns something about the aggregate state because his own option is good news about how many options exist in the economy.

3.3 Payoff function and aggregate information

The expected return to investment is $v(n)$, where $v(n)$ is increasing in n. Investment has positive prior value

$$V = \sum_{n=0}^{N} g(n)v(n) > 0,$$

but $v(n)$ can be negative for low values of n.

Interpretation. The investment decision is socially valuable on average but state-dependent. Learning the number of option holders matters because it shifts the posterior expected payoff.

3.4 Observable actions and information revelation

Investment is indivisible. When an agent invests, everyone observes the action and learns that the agent had an option. Failure to invest may also reveal information, depending on the equilibrium strategy.

Interpretation. Investment is both a payoff-relevant action and a signal. This dual role creates the strategic delay problem.

3.5 Period length and discounting

Time is discrete, with dates $t = 1, 2, \ldots$. Agents discount future payoffs by $0 < \delta < 1$. If the real period length is τ and the continuous discount rate is ρ, then

$$\delta = e^{-\rho\tau}.$$

Interpretation. The period length measures how fast agents can react to new information. The reaction speed is central to whether equilibrium generates delay or collapse.

4 Models and theoretical specifications

4.1 The outside-observer learning problem

Suppose each option holder independently invests with probability A. An outside observer sees k investments and updates beliefs about n. Define

$$V(k) = \sum_n g(n|k)v(n),$$

where $g(n|k)$ is the posterior after observing k investors. The observer's value of waiting is

$$W(A) = \sum_k p(k) \max\{V(k), 0\}.$$

Interpretation. Waiting has option value because the observer can condition investment on the number of early investors.

4.2 Option value of waiting

The option value of waiting is

$$W(A) - V.$$

It is positive only when the information revealed by early investment can change the action from invest to do not invest or vice versa.

Interpretation. This option value is the private reason to delay. The social problem is that waiting for information requires someone else to reveal it first.

4.3 Extensive-form timing game

There are N players, of whom n have investment options. A player with an option can invest at any date or never invest. Payoff from investing at date t is

$$\delta^{t-1}v(n),$$

and the payoff from never investing is zero. Players without options remain passive.

Interpretation. The game makes timing endogenous. Every active player trades off immediate investment against learning from other active players.

4.4 Strategies, beliefs, and histories

A symmetric strategy assigns an investment probability after each public history:

$$A : H \to [0, 1].$$

Beliefs are posterior probabilities about the number of option holders:

$$\mu(n|h).$$

Interpretation. In symmetric equilibrium, only the number of past investors matters, not their identities.

4.5 Symmetric perfect Bayesian equilibrium

A symmetric PBE consists of a strategy and beliefs such that each active agent optimizes at every information set and beliefs are Bayes-consistent along the equilibrium path.

Interpretation. The equilibrium is recursive: after each history, update beliefs, compute current investment value, compute the value of waiting, and determine whether agents invest, stop, or mix.

4.6 Equilibrium partition of beliefs

After a public history h, define current investment value $V(h)$ and waiting value $W(A(h), h)$. The equilibrium path has three regions:

- **Bad-belief region:** if $V(h) < 0$, no one invests and the game stops.
- **Good-belief region:** if immediate investment dominates even the best possible one-period learning, all active players invest.
- **Intermediate region:** active players mix with a unique probability $A(h)$ satisfying

$$V(h) = \delta W(A(h), h).$$

Interpretation. Mixing is the mechanism that produces gradual information revelation. The probability of early investment adjusts to make early movers indifferent between investing now and waiting.

5 Identification and theoretical design

5.1 What is being identified?

The paper does not estimate a causal parameter from data. It identifies a theoretical mechanism: pure information externalities can generate strategic delay, herd-like investment collapse, and underinvestment even when payoffs have no direct strategic complementarities.

Interpretation. The theoretical design separates information externalities from payoff externalities. This clean isolation is what makes the result powerful.

5.2 Key design choices

- **Pure informational externality:** other agents' actions affect beliefs, not payoffs.

- **Indivisible investment:** investment is a binary action, so information must be large enough to change a discrete decision.
- **Public actions:** investment is observable, so actions are signals.
- **Endogenous timing:** players choose when to reveal information through action.
- **Symmetry:** the equilibrium analysis focuses on symmetric PBE, the natural candidate when agents are ex ante identical.

5.3 Why period length matters

Let τ be the reaction lag. If periods are very short, agents can react almost instantly to early movers. But then the game also ends almost instantly because at least one investment must occur in every active period and there are finitely many option holders.

Interpretation. Faster reaction does not produce perfect information. Instead, it can produce faster collapse, because early absence of investment becomes self-confirming.

5.4 Why the large-number limit matters

As the number of players is replicated, the equilibrium investment probability per player falls so that the aggregate flow of investment remains finite. The limiting investment rate solves an option-value condition:

$$(1-\delta)V(m) = \delta\big[W^*(\beta, m) - V(m)\big].$$

Interpretation. A larger population does not automatically generate faster learning. More players simply create more agents who prefer to wait.

5.5 Main threats to interpretation

- **Symmetric equilibrium focus.** The analysis does not fully characterize asymmetric equilibria, and the paper notes that asymmetric equilibria can behave differently.
- **Binary actions.** The collapse result depends partly on the indivisibility of investment. Continuous actions can reveal small amounts of information for a long time.
- **Perfect observability of actions.** The baseline assumes actions are publicly observed. The paper discusses noise and argues that simple observational noise changes investment probabilities but not the basic amount of information revelation.
- **Stylized investment payoff.** The payoff is reduced form. The model is designed to study information revelation, not detailed firm investment technology.

6 Model-result boxes

6.1 Result Box 1: Proposition 1 and Proposition 2 — Value of information from early investment

Propositions 1–2
If the distribution of option holders is nondegenerate and investment probability is interior, the posterior value $V(k)$ is increasing in the number of observed investors. The observer invests only after enough investment is observed. The waiting value $W(A)$ increases in the amount of information revealed by the investment probability A.

Read. These propositions establish the option value of waiting. A higher number of early investors raises the posterior that the aggregate state is good. More informative early investment raises the value of delaying one's own decision.

Economic message. The same action that creates payoff also creates information. This is why the first movers are socially valuable but privately costly.

6.2 Result Box 2: Proposition 3 and Proposition 4 — Recursive characterization of equilibrium

> **Propositions 3–4**
> The one-step property holds along the equilibrium path. After any history, the game falls into one of three cases: no investment if beliefs are bad, immediate investment if beliefs are good, and unique mixed investment if beliefs are intermediate.

Read. The recursive structure means the equilibrium can be described by beliefs alone. Agents mix only when the value of current investment equals the discounted value of waiting for the next signal.

Economic message. Delay is not arbitrary. It is pinned down by indifference between revealing information today and learning from someone else tomorrow.

6.3 Result Box 3: Theorem 1 — Unique symmetric PBE outcome

> **Theorem 1**
> There exists a unique symmetric perfect Bayesian equilibrium outcome.

Read. Although dynamic games with incomplete information often have many equilibria, the symmetric outcome is pinned down by the three-region recursion in Proposition 4.

Economic message. The delay/collapse result is not a consequence of arbitrary equilibrium selection within the symmetric class.

6.4 Result Box 4: Proposition 5 and Proposition 6 — Period length and herd-like collapse

> **Propositions 5–6**
> Delay occurs if the discount factor is sufficiently high. No delay occurs if the discount factor is sufficiently low. In any symmetric PBE, all investment ends after at most N periods; as period length goes to zero, the real-time length of the investment process goes to zero.

Read. A high discount factor makes waiting attractive. But if periods become arbitrarily short, the finite number of potential investors implies the process ends almost immediately.

Economic message. Fast reaction can produce rapid information collapse, not perfect aggregation. This is the endogenous-timing counterpart of an informational cascade.

6.5 Result Box 5: Propositions 7–9 — Large economy and underinvestment

Propositions 7–9
In the large-player limit, the equilibrium rate of investment is bounded away from zero and infinity while beliefs remain intermediate. The limiting rate solves the option-value condition. The time profile becomes extreme: a period of negligible investment followed by either an investment surge or an investment collapse.

Read. Adding players does not make information flow arbitrarily fast. The aggregate rate adjusts so the option value of waiting equals its time cost.

Economic message. Large markets do not necessarily eliminate coordination failure. They can magnify the number of agents waiting for someone else to reveal information.

6.6 Result Box 6: Welfare and policy

Welfare discussion
In symmetric equilibrium, each player gets the same payoff as investing with no public information. First best would reveal n immediately. A temporary first-period subsidy to early investors can induce information revelation and approximate the first best.

Read. The policy is temporary because a permanent subsidy would not raise the marginal cost of delaying.

Economic message. The model gives a rationale for temporary pump-priming: early investment has a positive information externality that private agents do not internalize.

7 Short summary

- The paper builds a dynamic investment game with private information and endogenous timing.
- Agents privately know whether they have an investment option; exercising the option reveals information about aggregate prospects.
- Each agent wants others to invest first so he can learn before deciding.
- The equilibrium is inefficient because delay is costly and information is imperfectly revealed.
- The symmetric PBE outcome is unique and has a recursive three-region structure: invest, stop, or mix.
- Delay occurs when the value of waiting exceeds the immediate payoff.
- When periods are very short, the game ends quickly and can produce a herd-like collapse of investment.
- As the number of players grows, the information flow is governed by an option-value condition and does not speed up with the size of the market.
- The limiting time profile has negligible early investment followed by either a surge or collapse.
- The equilibrium has a bias toward underinvestment because collapse may occur even when investment is efficient.
- A temporary early-investment subsidy can help internalize the information externality and approximate first best.

8 Conclusion

8.1 Core conclusion

Chamley and Gale show that strategic delay can arise purely from informational externalities. Even when an agent's payoff is independent of other agents' actions, actions reveal private information. This induces agents to wait for others, but if too many wait, information stops being produced.

8.2 Economic intuition

Information is a public good in this model. The first investor bears the cost of acting under uncertainty, while later agents learn from the first investor's action. Private incentives therefore underprovide early action. The equilibrium must leave some inefficiency: either investment occurs too slowly, or the market wrongly infers bad news from the absence of investment and collapses.

8.3 Incremental contribution in one sentence

The paper transforms the cascade/herding idea from an exogenous-order learning model into an endogenous timing model and shows that the attempt to learn from others can itself determine whether information is revealed, delayed, or destroyed.

8.4 Policy implication

The welfare discussion suggests that temporary subsidies to early investment can be useful precisely because they compensate early movers for the information they generate. The subsidy should be temporary: a permanent subsidy changes levels but does not necessarily make delay more costly at the margin.

8.5 Limitations and scope

- The analysis focuses on symmetric equilibria.
- Investment is binary and indivisible.
- The model abstracts from direct payoff complementarities, financing constraints, and rich firm heterogeneity.
- The welfare policy is sketched rather than fully embedded in a general equilibrium fiscal framework.

8.6 Takeaway

The model explains how a recession-like investment freeze can occur even without direct strategic complementarities. If everyone waits for information from everyone else, the economy can stop revealing the very information needed to justify investment.